Date
Page

Assignment:-1

1) Explain about connection-oriented network services.

A) Connection-Oriented networservice is a type of communication where a connection is first established between the sender and receiver before any actual data transfer takes place. This connection ensures that data is transmitted in an organized and reliable manner.

The most common example of a connection oriented service is Transmission Control Protocol (TCP). In TCP, a three way handshake process is used to establish the connection. First, the sender sends a SYN message, the receiver replies with a SYN-ACK, and then the sender replies with an ACK. Once this handshake is complete, both parties can begin exchanging data.

2) Difference between web-server and proxy server.

A) A Web Server and proxy server serve different purpose within a network, though both deal with web-based communications.

A Web server is a computer system that hosts websites. It stores, processes, and delivers web pages to clients using the HTTP or HTTPs protocol. When a user types a website URL in their browser, the request goes to the web server, which responds by sending back the requested webpage. Popular web server software includes Apache, Nginx & Microsoft IIS.

On the other hand, a Proxy Server acts as an intermediary between the client and the web server. When a client makes a request for a resource, it goes to the proxy server first. The proxy can then forward the request to the destination webserver or respond from its own cache.

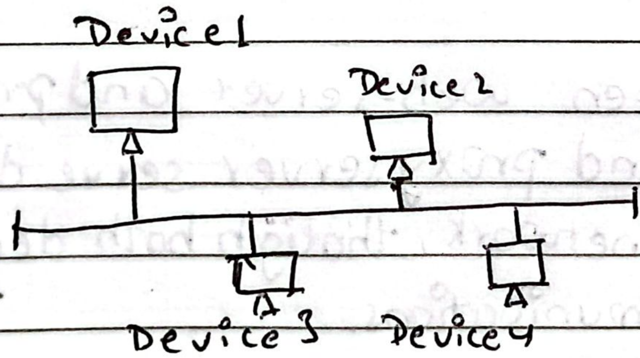
3) Explain different types of network topologies.

↳ Network topology refers to the physical or logical layout of nodes and connections in a network.

There are several types of topologies:

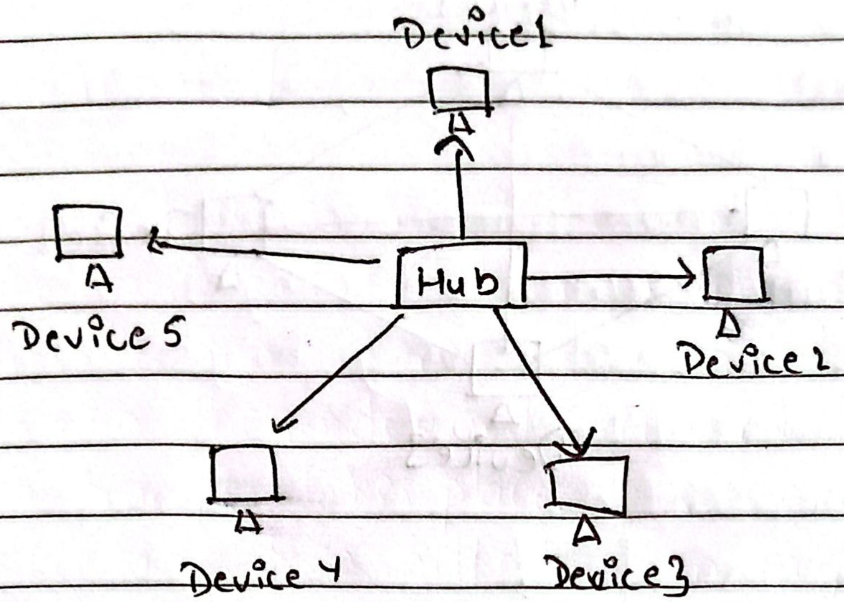
1) Bus Topology:

↳ All devices are connected to a single central cable. It's easy to set up & requires less cable. A failure in the main cable causes the entire network to go down and data collisions are common.

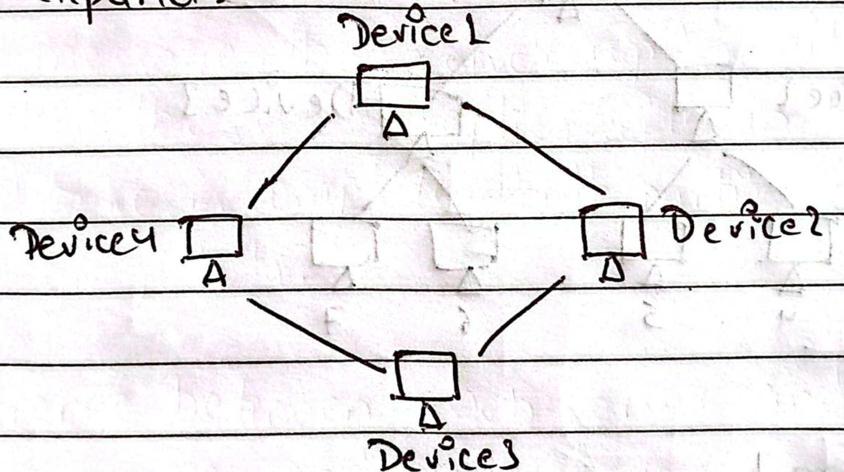


2) Star Topology

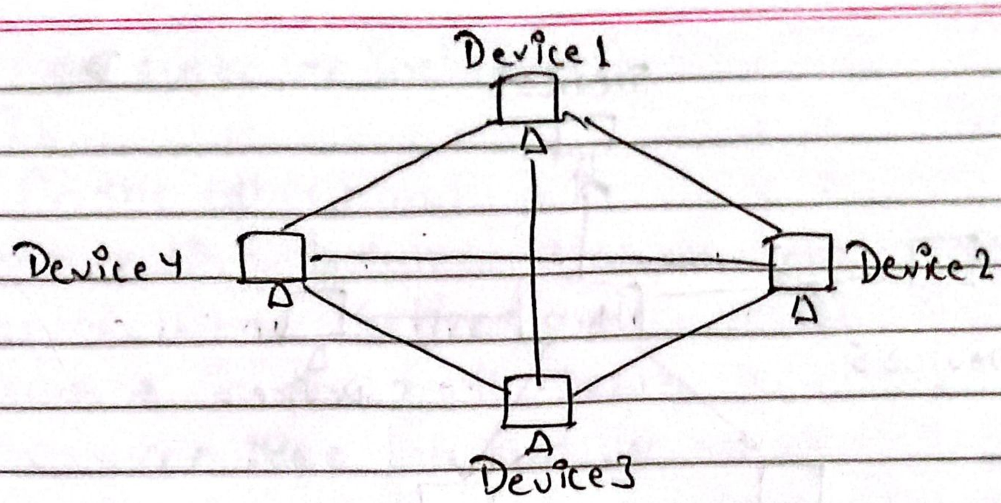
↳ All devices are connected to a central hub or switch. It's easy to manage and expand. If one node fails, it doesn't affect the others but if the hub fails, the whole network goes down



Ring Topology
Devices are connected in a circular fashion. Data travels in one direction. It's better than a bus in avoiding collisions but harder to troubleshoot and expand.

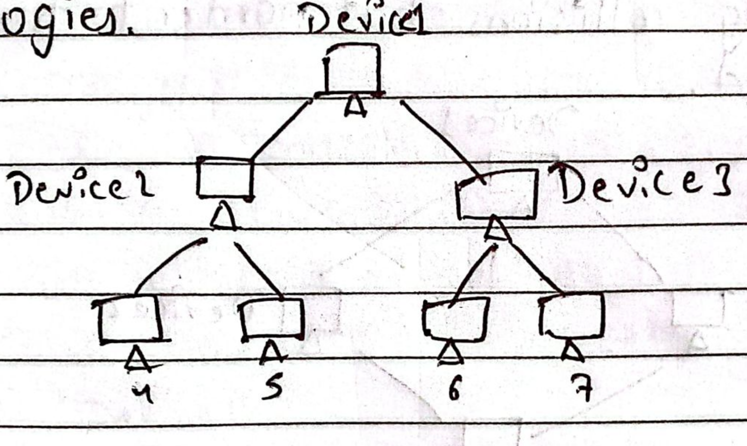


Mesh Topology
Every node is connected to every other node. It provides high reliability and redundancy. However, its expensive ~~and easy management but the same drawbacks as both its parent topologies.~~ due to number of connections needed.



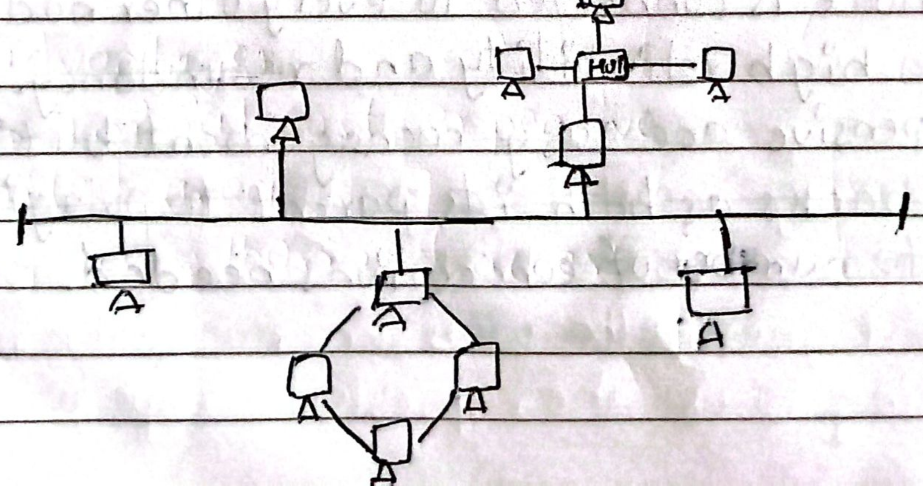
5) Tree Topology

↳ A combination of star and bus topologies. It allows for expansion and easy management but has the same drawbacks as both it's parent topologies.



(6) Hybrid Topology

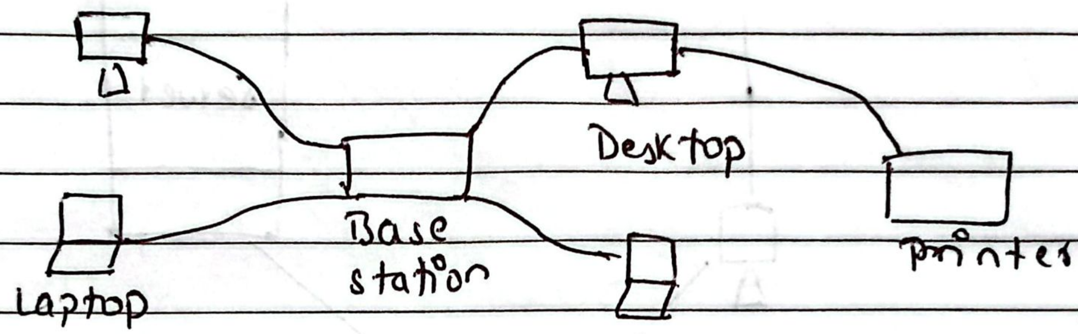
↳ Combines two or more topologies. It is scalable and flexible but complex and costly.



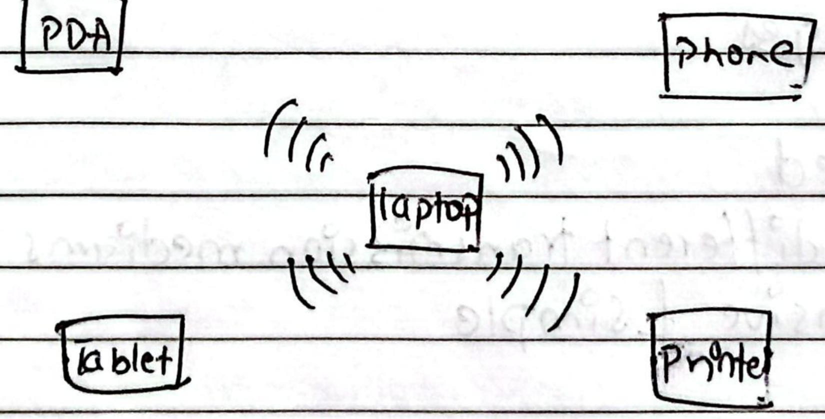
(H) Explain different types of computer network ~~types~~.
→ The different types of computer network are:-

(i) Personal Area Network (PAN)
→ PAN is the most basic type of computer network. It is a type of network designed to connect devices within a short range, typically around one person. It offers a network range of 1 to 100 meters from person to device providing communication.

Types
(a) Wired PAN



(b) Wireless PAN

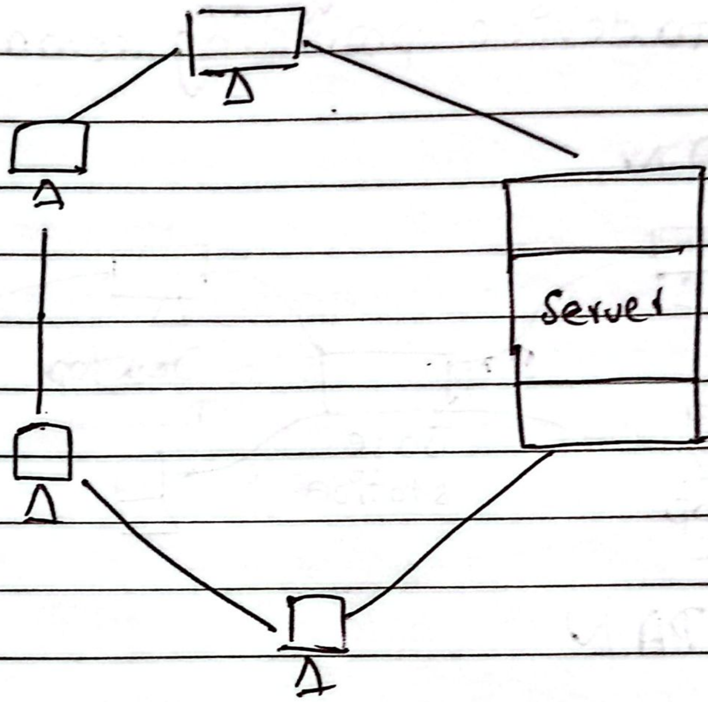


Advantages
- Flexible and high efficient
- Easy and Portable
- Easy setup & low cost

Disadvantages
- Not Compatible
- low network coverage area

② Local Area Network (LAN)

↳ LAN is the most frequenty used network. It is a computer network that connects computers through a common communication path, contained within a limited area. It ranges upto 2 km & transmission speed is very high with easy maintenance & low cost.



Advantages

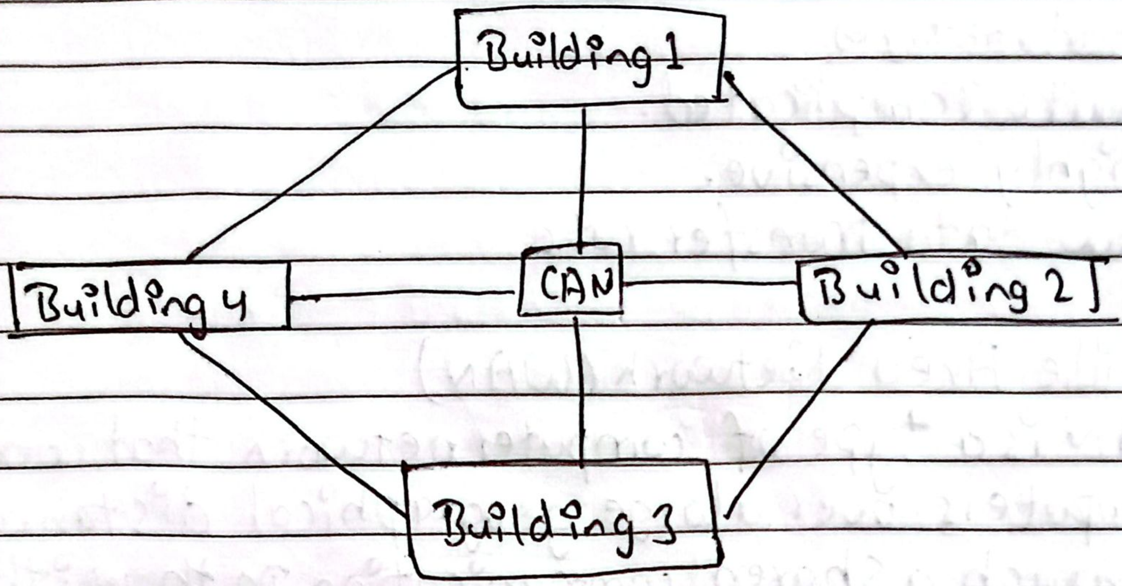
- Privacy
- Highspeed
- Supports different transmission mediums
- Inexpensive & simple

Disadvantages

- ~~Expe~~ Initial cost expensive
- limited area
- Easy unauthorized access

③ Campus Area Network (CAN)

↳ CAN is a type of computer network that is usually used in places like a school or colleges. This network covers a limited geographical area with range from 1km to 5km.

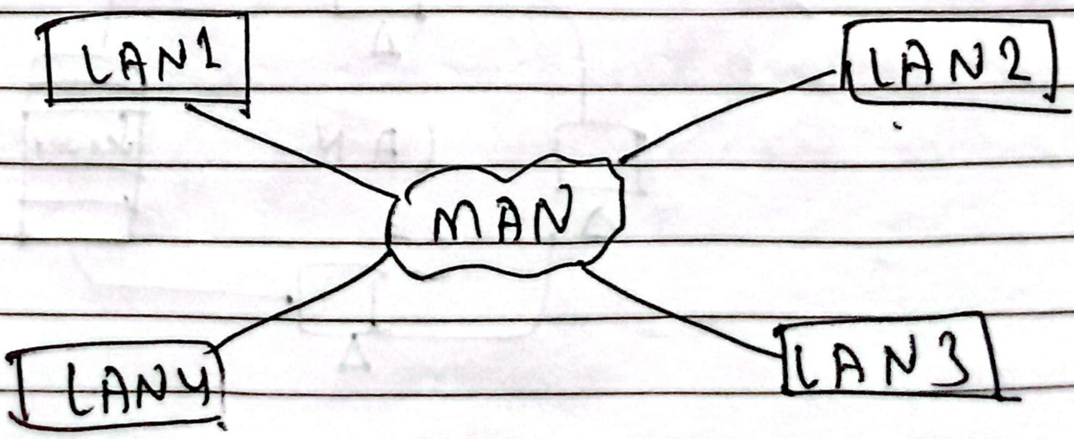


Advantages

- ~~Speed~~ High speed
- More secured
- Cost effective.

④ Metropolitan Area Network (MAN)

↳ MAN is the type of computer network that connects computers over a geographical distance through a shared communication path over a city or town with range from 5km to 50km.



Advantages

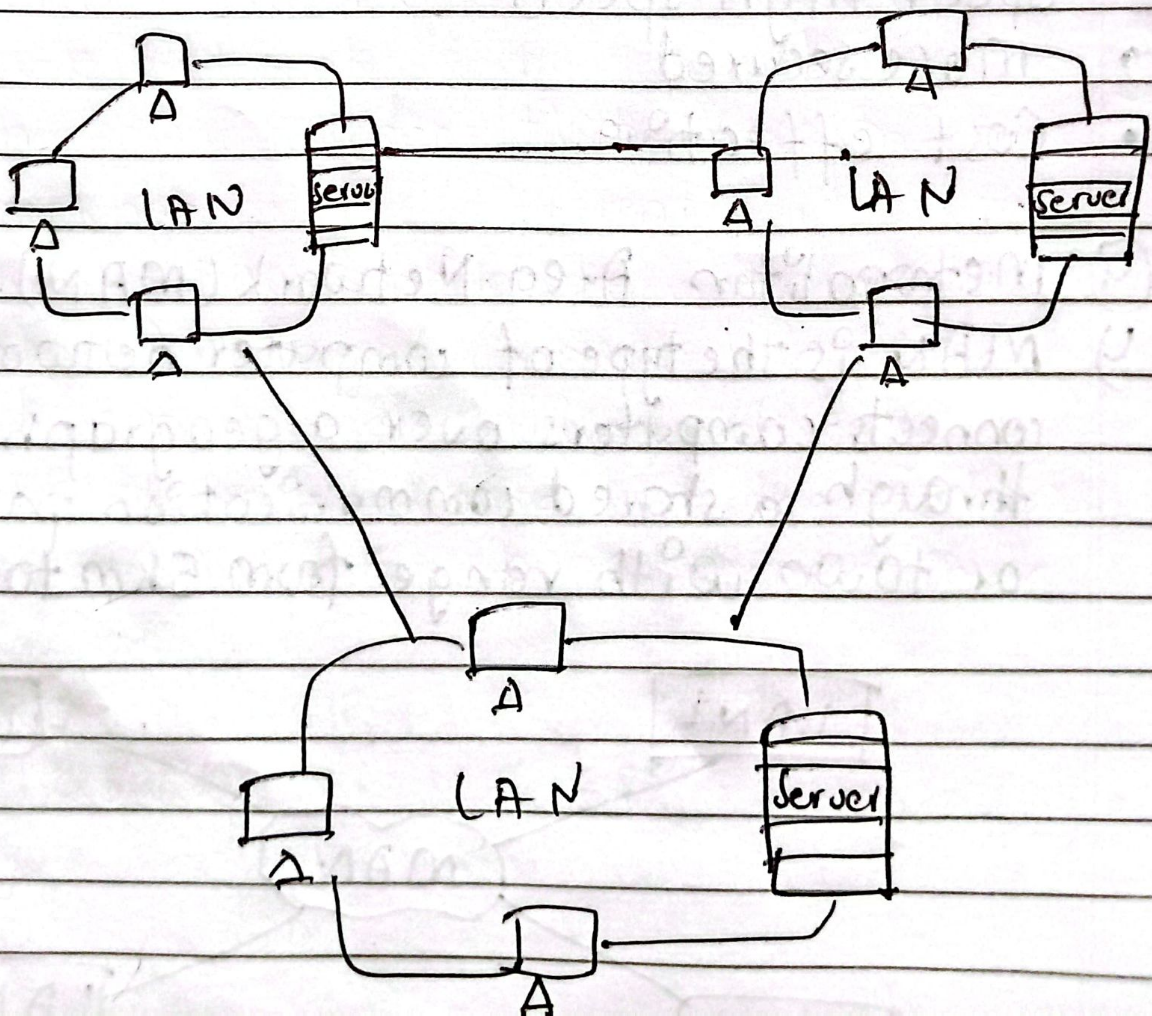
- Highspeed
- Moresecured & strict.
- Bothdirection transmission concurrently.
- Multiple users at a time.

Disadvantages

- Quite complicated.
- Highly expensive.
- Low data transfer rate.

(5) Wide Area Network (WAN)

→ WAN is a type of computer network that connects computers over a large geographical distance through a shared communication path with a range above 50km.



Date
Page

Advantages
- Transmit data quickly and cheaply.
- Data can be stored in centralised manner.

Disadvantages
- High traffic congestion
- Noise and error
- less fault tolerance ability.

(5) Write Short Notes:

a) Protocol and standards
i) Protocol and standards are important in computer networks. They are like the rules & guidelines that allow different devices & systems to communicate and systems to communicate and work together smoothly. Protocols define how data is sent, received & processed while standards ensure that various technologies are compatible with each other. This coordination is critical for the internet and other networks to function constanly and efficiently.

A protocol is a set of rules that determines how data is sent & received over a network. The protocol is just like a language that computers use to talk to each other, ensuring they understand and can respond to each other's messages correctly.

Types of Protocol

- Network layer Protocol
- Transport " "
- Application " "
- Wireless Protocol
- Routing "
- Security "
- Internet Protocol

Standards are the set of rules for data commu-nication that are needed for the exchange of information among devices.

Types of Standards.

- Defacto Standard
- De Jure Standard.

b) Backbone Network.

4) Backbone Network is a network containing a high capacity connectivity infrastructure that backbone to the different part of the network. Actually a backbone network allows multiple LANs to get connected in a backbone network, not a single station is directy connected to the backbone but the stations are part of LAN, & backbone connect those LANs.

Drawbacks

1) Poor Reliability
2) Capacity
3) Cost

Types of backbone network

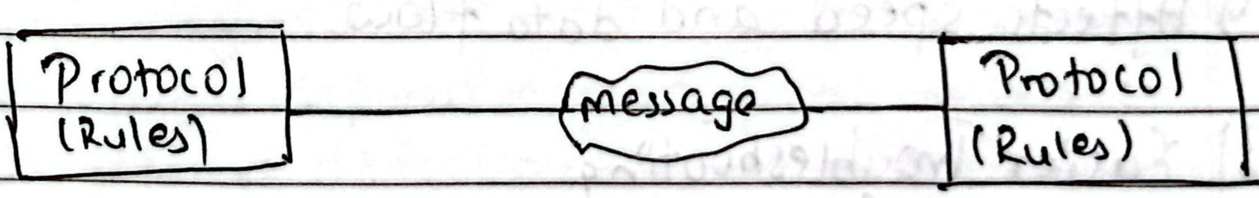
- Serial backbone
- Distributed backbone
- Collapsed backbone
- Parallel backbone

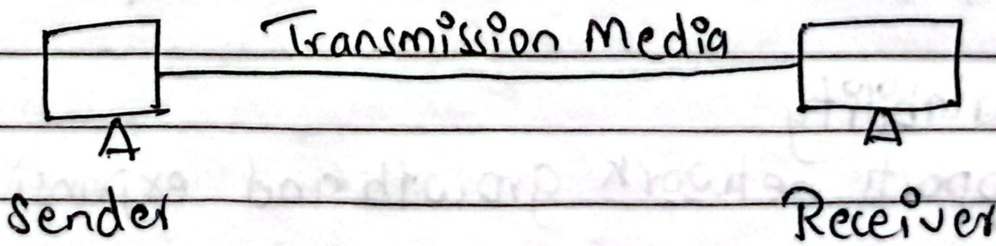
Functions

- Aggregation
- Authentication
- Call control and switching
- Charging
- Service invocation
- Gateways.

(6) Define Protocol. Why do we need standards?

→ A protocol is a set of rules that determines how data is sent and received over a network. The protocol is just like a language that computers use to talk to each other, ensuring they understand and can respond to each other's messages correctly.





We need Standards becauses of
1) Interoperability → Devices from different vendors can work together.
2) Compatibility → Ensures smooth communication using common protocols.
3) Scalability → Easier to expand and upgrade networks
4) Security → Standard protocols ensure reliable and tested security.
5) Cost - effectiveness → Reduces costs
6) Testing and Certification → Enables quality assurance through standard tests.

7) Why do we need network topology ?. Explain startopology along with its merits and demerits.
↳ We need network topology because of
1) Efficient Design
↳ Helps plan how devices connect.

2) Performance Optimization
↳ Affects speed and data flow.

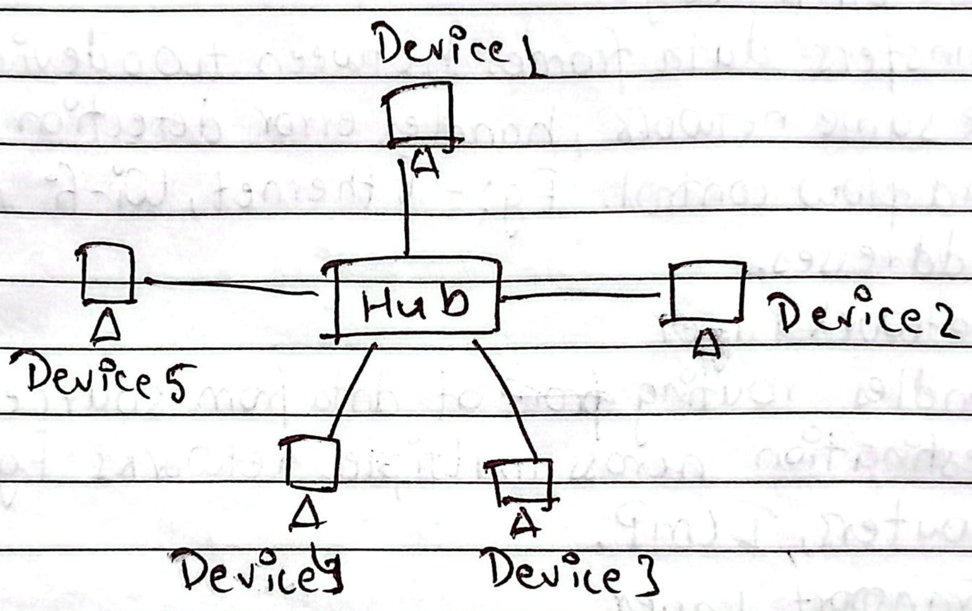
3) Easier Troubleshooting
↳ Simplifies fault detection and repair.

4) Scalability
↳ Supports network growth and expansion.

5) Cost Control
↳ ~~Suppo~~ Impacts hardware & setup costs.

6) Improve Management
→ Aids in monitoring & maintaining the network.

→ Star topology is a network setup in which each device is connected to a central node called hub. If one device wants to send data to another device, it has to first send the information to hub & hub transmits that data to required device.



Merits
1) It is very reliable and high performing
2) It is less expensive and no disruptions.
3) Easy fault detection because links are easily identified.

Demirits.
1) Requires more cable
2) More expesive than bus topology
3) If hub goes down everything goes down.

(8) What is protocol? Explain each layer of OSI model indetail.

→ A protocol is a set of rules that define how data is transmitted and received over a network.

Layers of OSI model

1) Physical layer

→ Transmits raw bits (0s & 1s) over a physical medium. Eg:- Ethernet cables, fiber optics, radio signals.

2) Data Link layer

→ Transfers data frames between two devices on the same network, handles error detection / correction and flow control. Eg:- Ethernet, Wi-fi, MAC addresses.

3) Network Layer

→ Handles routing ~~from~~ of data from source to destination across multiple networks. Eg: IP, routers, ICMP.

4) Transport Layer

→ Provides reliable data delievery, error recovery and flow control between devices. Eg: Ensuring a file downloads completely & in order.

5) Session Layer

→ Manages sessions between applications, including starting, maintaining and ending sessions. Eg: Remote procedure call, login sessions, video call sessions. Eg: Remote procedure call, login sessions, video call sessions.

Date
Page

6) Presentation layer
↳ Translates, encrypts and compresses data for the application layer. Eg: Converting a video file to a playable format.

7) Application layer
↳ Closest to the user; provides network services to applications. Eg:- HTTP, SMTP, FTP etc.

(Q) Explain each layer of TCP/IP model in detail. Compare it with OSI model.

↳ Layers of TCP/IP model

1) Network Interface layer
↳ Handles physical transmission of data on the local network. Eg:- Ethernet, Wi-fi, ARP.

2) Internet layer
↳ Delivers packets across networks using IP addressing. Eg:- IP, ICMP, ARP, IGMP.

3) Transport layer
↳ Ensures reliable or fast data transfer between hosts. Eg: TCP - ~~Reliable~~ → file download
UDP → live streaming video.

4) Application layer
↳ Provides services for applications to access the network. Eg: HTTP, FTP, SMTP, DNS.

Page

Comparing it with OSI model

1. Number of layers
   - OSI : 7 layers
   - TCP/IP : 4 layers

2. Layer functions
   - OSI → ~~the~~ strictly separated
   - TCP/IP → merges into a single

3. Usage
   - OSI → ~~theoot~~ theoretical model
   - TCP/IP → Practical model

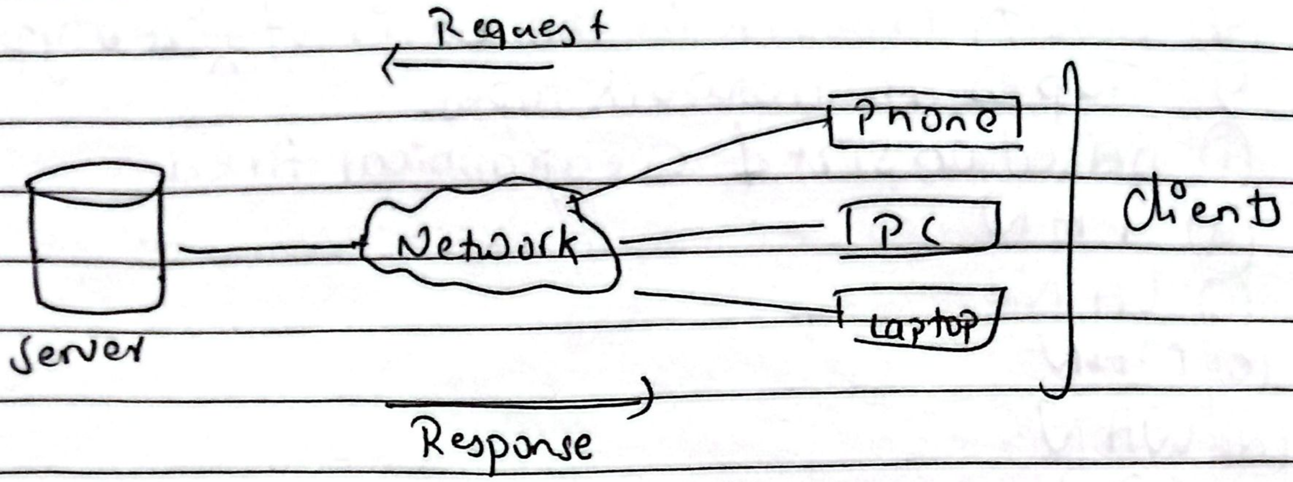
4. Protocol Binding
   - OSI → Protocol independent
   - TCP/IP → Protocol-specific

10) What are the two network architecture in practice explain with example?

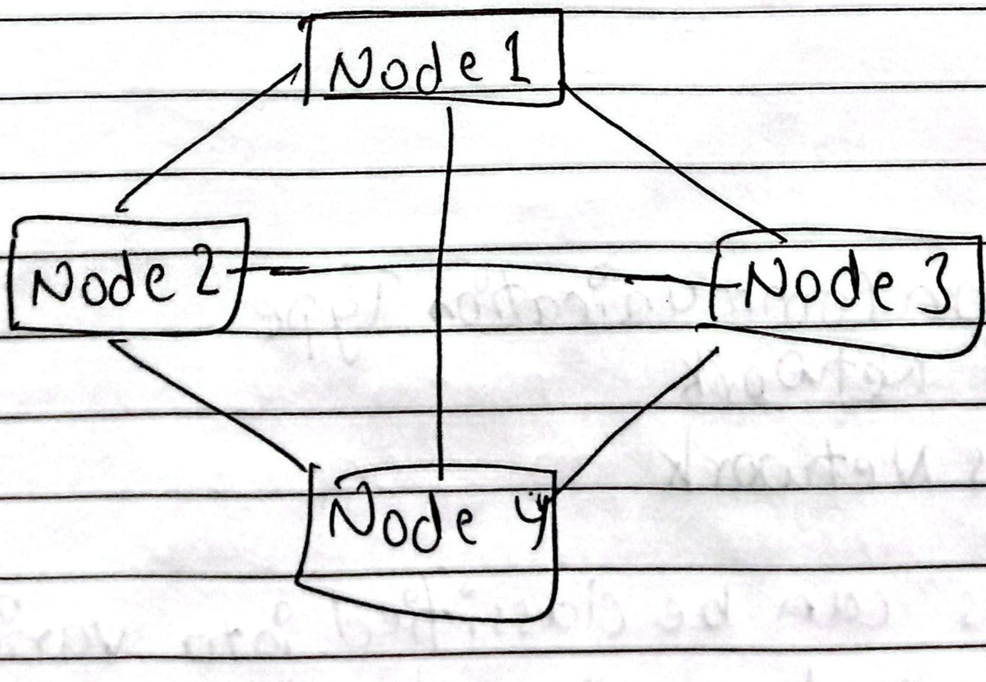
→ The two network architecture in practice are:-

1) Client-Server Architecture

→ The client-server architecture is a centralized model where multiple client devices connect to a central server to request and access services or resources. The server hosts, manages, and delivers data or services, such as websites, emails or files while the clients acts as consumers of these resources. Example: Web, email, banking apps.



Peer-to-Peer (P2P) Architecture:
In peer-to-peer architecture, every device in the network acts as both a client & a server. Instead of relying on a central server, peers communicate directly with each other to share resources, such as files or processing power. This model is decentralized, meaning there's no single point of control or failure.
Eg:- Torrent sharing, blockchain.



(4) Classify network in the basis of ~~size~~ network & types

y Types of network types.

(1) Based on Size & Geographical Area

(a) PAN
(b) LAN
(c) MAN
(d) WAN
(e) GAN

(2) Based on Network Architecture

(a) Client-Server
(b) Peer to Peer
(c) Hybrid.

(3) Based on Network Topology

(a) Bus
(b) Star
(c) Ring
(d) Mesh
(e) Tree
(f) Hybrid

(4) Based on Communication Type

(a) Wired Network
(b) Wireless Network

Networks can be classified into various types based on size, structure, architecture & communication methods.

Page____

Based on size & geographical coverage, we have PANs, LANs, MANs, WANs & GANs.
From an architectural point of view, networks are either client-server or peer-to-peer (P2P) & hybrid architecture is combination of both.
Networks can also be categorized by their topology or physical layout including, bus, star, mess, tree & hybrid. lastly, based on how devices communicate, networks can be wired or wireless.
Each types of network ~~services~~ serves specific purposes depending on factors like range, performance, cost and use-case requirements.

(12) Explain client/server network. How is it different from peer to peer network?

→ A client-server network is a type of network architecture where multiple devices called clients are connected to a central server that provides services, resources or data.
In this model, the server is a powerful computer that manages files, applications, network traffic, security & more while the clients send requests to the server & receive responses.

It is different from peer to peer network because of

1. Client-server is centralized whereas peer to peer is decentralized.
2. It has server controls but not in peer to peer network.

Page

(3) Client request data from server in client-server ~~to~~ whereas peers share data directly in peer-to-peer.
(4) Client-server network is more expensive than peer-to-peer.
(5) It is also more faster & reliable than other.
(6) More secured than peer to peer server.

(13) Explain LAN with example. How is it different from PAN?
→ A LAN (Local Area Network) is a type of computer network that connects devices within a limited ~~grap~~ geographical area such as home, office, school or building. It ranges upto 2km.

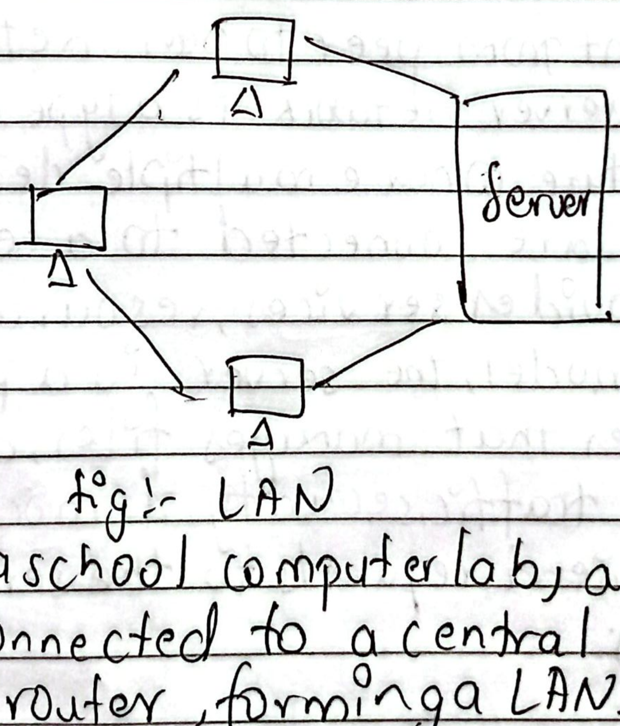


fig:- LAN

Eg:- In a school computer lab, all the computers are connected to a central switch or Wi-fi router, forming a LAN.

It is different from PAN because of:-
(1) Covers a larger area than as compared to PAN.
(2) LAN connects multiple devices whereas

PAN only connects personal devices.

3. LAN uses Ethernet cables or Wifi for connection but PAN uses bluetooth, USB or infrared for connection.
4. Higher speed & data transfer rate compared to PAN.
5. Manged by network administrators but managed by the individual user in PAN.